Nguyen Hoang Anh Duy

Data Analyst Intern candidate · Ho Chi Minh City, Vietnam
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Summary

Final-year Data Science student (HCMUS, GPA 3.62/4.00) with SQL, Python and Power BI skills shown in two end-to-end projects: a PostgreSQL warehouse and Power BI dashboard on 96,477 e-commerce orders, and a shop-level growth analysis of **Home Credit Vietnam**'s 11,752 partner shops that won **Business Intelligence Season 10** as team leader. Seeking a full-time Data Analyst Intern role in Ho Chi Minh City.

Technical Skills

Languages: SQL (PostgreSQL, DuckDB), Python (pandas, NumPy, scikit-learn)
Visualization & BI: Power BI (DAX, Power Query, data modeling, drill-through), Excel
Data Work: Data cleaning and validation, reconciliation tests, star schema, ELT pipelines
Analysis Methods: RFM and K-Means segmentation, survival analysis, cohort analysis, peer benchmarking
Workflow: Git, GitHub, Jupyter Notebook

Projects

Business Intelligence Season 10 (ITB Club, UEL) — Champion *Sep – Oct 2026*
Python, DuckDB, SQL | Team leader of 5 | github.com/nhaduy-data/bi10-shop-growth-potential

- Built the team's shared cleaning pipeline over **2 GB** of raw files in Python and DuckDB SQL, flagging **36% of leads** with invalid employee IDs before any analysis began
- Analyzed shop-level contract history for **11,752 partner shops** and found that growth came from opening new shops, while each shop stayed at about 8 applications/month and **33.8% of active shops** (3,975) signed no contract in Q3 2026
- Modeled attainable potential by benchmarking each shop against the top 25% of similar shops, estimating **16,300 more contracts/month (+41.6%)**, 75% of it from sales-agent headcount and productivity
- Recommended one action per shop (invest, pilot, hold or fix risk) and presented it to a judging panel that included Home Credit Vietnam's Head of Sales Planning

Customer Lifecycle Analytics — Olist E-commerce *Jun – Sep 2026*
PostgreSQL, Python, Power BI | github.com/nhaduy-data/customer-lifecycle-analytics

- Built a **PostgreSQL** star-schema warehouse (96,477 orders) with SQL reconciliation tests, catching a join fan-out that **doubled revenue** on multi-item orders and split orders that inflated the repeat-purchase rate from 2.16% to 3.0%
- Segmented 93,357 customers with RFM and K-Means in Python (scikit-learn), finding **39.3% of revenue** in an At Risk group that rule-based RFM scores could not separate from Potential Loyalists
- Set churn thresholds at 175 / 288 days with survival analysis in Python; the common 90-day rule would have **misclassified 45%** of customers who later returned
- Designed a 5-page **Power BI** dashboard (DAX, Power Query, drill-through, seller watch-list) showing late delivery hurts ratings (**2.3 vs. 4.3 stars**) far more than repeat purchases (2.2% vs. 1.6%), so the fix belongs to logistics, not retention campaigns

Education

University of Science, Vietnam National University HCMC Ho Chi Minh City
Bachelor of Science in Data Science — ***GPA: 3.62/4.00*** *2023 – 2027 (Expected)*

- **Relevant Coursework:** Database Management Systems (10/10), Data Visualization (9.8/10), Data Mining (9.3/10), Python for Data Science (9.2/10)

Certifications & Languages

Google Data Analytics Professional Certificate — Coursera, 2025
English: Professional working proficiency (comfortable reading technical documentation); TOEIC in preparation